

KUO-HAN HUNG

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🏠 [khhung-906.github.io](https://github.com/khhung-906)

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Education

Stanford University

June 2027 (Expected)

M.S. in Computer Science

Stanford, CA

- GPA: 4.0/4.0; Specialization: Artificial Intelligence and Robotics.

National Taiwan University

Dec. 2024

B.S. in Computer Science and Information Engineering

Taipei, Taiwan

- GPA: 4.24/4.3, Top 3% of Class. **Awards:** Bachelor Thesis Dean's Award, Presidential Award (×2), Dean's List (×3).

Technical Skills

- **Robotics & Systems:** Real-Robot Deployment and Training, Real-to-Sim, Real-Time Control, ROS, Docker, Linux.
- **Parallel & Languages:** Distributed Training, Ray, SLURM, GCP, Python, C/C++, Bash, PyTorch, JAX, NumPy.

Research Experience

Stanford Artificial Intelligence Laboratory (ILIAD & IRIS Labs)

Sept. 2025 – Present

Graduate Researcher, Advisors: Prof. Dorsa Sadigh & Prof. Chelsea Finn

Stanford, CA

- **RL finetuning of real-time VLA policies** under inference latency. Decoupled slow VLA proposals from a fast edit policy, raising success from 42% to 97% across six tasks including dexterous light-string routing. [1]
- **Sample-efficient RL finetuning of pretrained VLAs.** Built a stable system and algorithm to improve VLA with residual action edits and Q-guided sampling, reaching 30/30 success with 19 min of online data (100+ GitHub stars). [2]
- **Scaling Q-functions** with Transformers in offline RL. Introduced attention entropy regularization to stabilize and scale Transformer value functions, gaining 43% as model size grows where prior methods fail. [3]
- **Visual representation pretraining** for reinforcement learning (ongoing).

Robot Learning Lab

Oct. 2024 – May 2025

Undergraduate Researcher, Advisor: Prof. Shao-Hua Sun

Taipei, Taiwan

- **Manipulation policies learned from action-free videos**, using optical flow as an action surrogate. [5]

Communication and Multimedia Lab

May 2022 – May 2024

Undergraduate Researcher, Advisors: Prof. Winston H. Hsu & Prof. Yi-Ting Chen

Taipei, Taiwan

- **Vision-instruction correlation rewards** for long-horizon manipulation: 43% over prior best. [4]
- **Behavior error detection** in few-shot imitation policies via representation patterns.

Work Experience

IBM Research

June 2024 – Sept. 2024

Research Intern, Host: Dr. Pin-Yu Chen

Yorktown Heights, NY

- Found that attention drifts from the original instruction during **prompt injection attacks on LLM agents**.
- Traced it to specific attention heads and built a detector with a 31% AUROC gain over prior best.

Microsoft

July 2023 – June 2024

Applied Scientist Intern, Host: Dr. Chiquan Zhang

Mountain View, CA (Remote)

- **Integrated Bing Maps with Microsoft Copilot** for conversational, multi-destination map queries.
- Formulated a pathfinding algorithm that learns from history, **achieving a 10× speedup over Dijkstra**.

Selected Publications

(* indicates equal contribution)

- [1] **Reinforcement Learning for Real-Time Vision-Language-Action Policies**
Perry Dong*, Kuo-Han Hung*, Dorsa Sadigh, and Chelsea Finn. Under Review
- [2] **EXPO-FT: Sample-Efficient Reinforcement Learning Finetuning for Vision-Language-Action Models**
Perry Dong*, Kuo-Han Hung*, Tian Gao, Dorsa Sadigh, and Chelsea Finn. **CoRL** 2026
- [3] **TQL: Scaling Q-Functions with Transformers by Preventing Attention Collapse**
Perry Dong*, Kuo-Han Hung*, Alexander Swerdlow, Dorsa Sadigh, and Chelsea Finn. **ICML** 2026
- [4] **VICtoR: Learning Hierarchical Vision-Instruction Rewards for Long-horizon Manipulation**
Kuo-Han Hung*, Pang-Chi Lo*, Jia-Fong Yeh*, Han-Yuan Hsu, Yi-Ting Chen, and Winston Hsu. **ICLR** 2025
- [5] **Learning Skill Abstraction from Action-Free Videos via Optical Flow**
Hung-Chieh Fang*, Kuo-Han Hung*, ..., Min-Hung Chen, and Shao-Hua Sun. **ICML Workshop** 2025